

Applied Mathematics for Chemists

(화학수학)

Lecture 11

Differential Equations

Terminology

- Order of DE (계수): order of the highest derivative
- Linear (선형) vs. nonlinear (비선형)
 - Linear: contains the function (y) and its derivatives only to the first power (Note: terms in x can be nonlinear)

$$a_n(x) \frac{d^n y}{dx^n} + a_{n-1}(x) \frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_1(x) \frac{dy}{dx} + a_0(x)y = g(x)$$

- Nonlinear: otherwise

$$(1 - y) \frac{dy}{dx} + 2y = e^x, \quad \frac{d^2 y}{dx^2} + \sin y = 0, \quad \dots$$

Linear Differential Equations

$$a_n(x) \frac{d^n y}{dx^n} + a_{n-1}(x) \frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_1(x) \frac{dy}{dx} + a_0(x)y = g(x)$$

- Homogeneous (동차): $g(x) = 0$
- Inhomogeneous (비동차): $g(x) \neq 0$

Higher-Order ODEs: Linear Case

$$a_n(x) \frac{d^n y}{dx^n} + a_{n-1}(x) \frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_1(x) \frac{dy}{dx} + a_0(x)y = g(x)$$

- Superposition principle (§ 12.2.1)
- Reduction of order (not in Mortimer)
- Linear ODEs with constant coefficients (§ 12.2, 12.3)
 - Homogeneous (§ 12.2)
 - Inhomogeneous (§ 12.3)
- We will not discuss nonlinear ODEs

Homogeneous Linear Equations

$$a_n(x) \frac{d^n y}{dx^n} + a_{n-1}(x) \frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_1(x) \frac{dy}{dx} + a_0(x)y = 0$$

- Superposition principle (중첩 원리):

If $y_1(x), y_2(x), \dots, y_k(x)$ are the solutions of the ODE, the linear combination

$$y = c_1 y_1(x) + c_2 y_2(x) + \cdots + c_k y_k(x)$$

is also a solution of the ODE for arbitrary constants $c_1, c_2, \dots, c_k$.

Reduction of Order

$$a_2(x) \frac{d^2 y}{dx^2} + a_1(x) \frac{dy}{dx} + a_0(x)y = 0$$

If a solution y_1 is known,
we can obtain the other solution y_2

$$y_2(x) = y_1 \int \frac{1}{y_1^2} e^{-\int P dx} dx$$

Linear ODEs with Constant Coefficients

- Homogeneous

$$a \frac{d^2 y}{dx^2} + b \frac{dy}{dx} + cy = 0$$

- Inhomogeneous

$$a \frac{d^2 y}{dx^2} + b \frac{dy}{dx} + cy = g(x)$$

Homogeneous Case

$$a \frac{d^2 y}{dx^2} + b \frac{dy}{dx} + cy = 0$$

- Trial solution (시행해): $y = e^{\lambda x}$

$$a\lambda^2 e^{\lambda x} + b\lambda e^{\lambda x} + ce^{\lambda x} = 0$$

- Characteristic equation (고유방정식)

$$a\lambda^2 + b\lambda + c = 0$$

Three possible cases!

(1) Distinct Real Roots

$$\lambda_1 = \frac{-b + \sqrt{b^2 - 4ac}}{2a}, \quad \lambda_2 = \frac{-b - \sqrt{b^2 - 4ac}}{2a}$$

We find two solutions: $y_1 = e^{\lambda_1 x}$ and $y_2 = e^{\lambda_2 x}$

The general solution is

$$y = c_1 y_1 + c_2 y_2 = c_1 e^{\lambda_1 x} + c_2 e^{\lambda_2 x}$$

(2) Repeated Real Roots

$$\lambda_1 = \lambda_2 = \frac{-b}{2a}$$

We obtain one solution: $y_1 = e^{\lambda_1 x}$

Using the reduction of order,

$$y_2 = y_1 \int \frac{e^{-\int P dx}}{y_1^2} dx = e^{\lambda_1 x} \int \frac{e^{2\lambda_1 x}}{e^{2\lambda_1 x}} dx = x e^{\lambda_1 x}$$

$$y = c_1 y_1 + c_2 y_2 = c_1 e^{\lambda_1 x} + c_2 x e^{\lambda_1 x}$$

(3) Conjugate Complex Roots

$$\lambda_1 = \frac{-b + \sqrt{b^2 - 4ac}}{2a} = \alpha + i\beta$$

$$\lambda_2 = \frac{-b - \sqrt{b^2 - 4ac}}{2a} = \alpha - i\beta$$

We find two solutions: $y_1 = e^{\lambda_1 x}$ and $y_2 = e^{\lambda_2 x}$

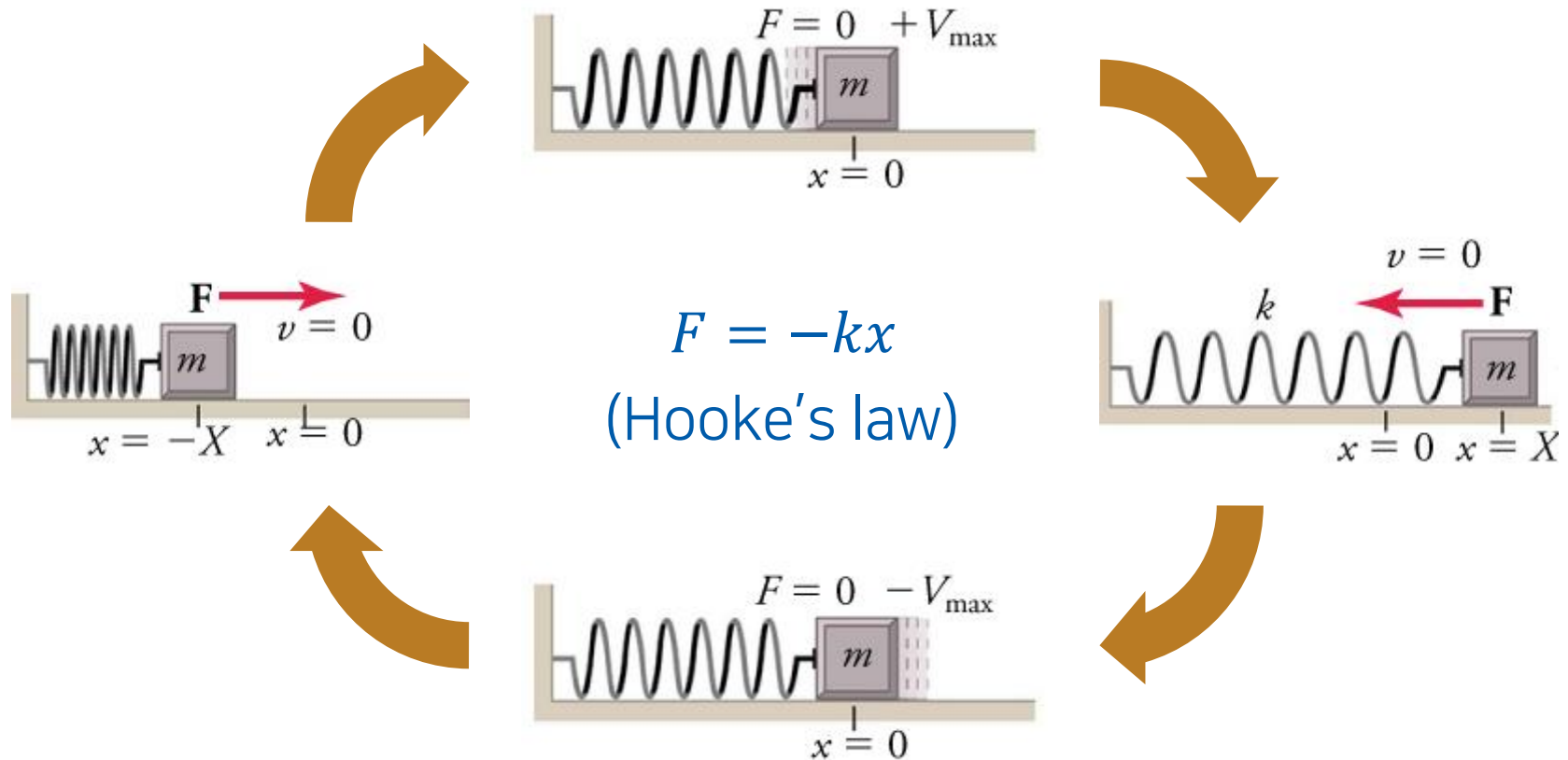
The general solution is

$$y = e^{\alpha x} (c_1 \cos \beta x + c_2 \sin \beta x)$$

Today's Class

- Homogeneous linear ODEs with constant coefficients
 - Application 1: ideal harmonic oscillator
 - Application 2: damped harmonic oscillator
- Inhomogeneous linear ODEs with constant coefficients
 - Complementary function
 - Method of undetermined coefficients

Application: Harmonic Oscillator (조화진동자)



(Image: <https://openstax.org/books/physics/pages/5-5-simple-harmonic-motion>)

Equation of Motion (운동방정식)

- Newton's second law

$$F = ma = m \frac{dv}{dt} = m \frac{d^2x}{dt^2}$$

- Equation of motion

$$F = m \frac{d^2x}{dt^2} = -kx$$

$$\frac{d^2x}{dt^2} + \frac{k}{m}x = 0$$

Solving Equation of Motion

$$\frac{d^2x}{dt^2} + \frac{k}{m}x = 0$$

The characteristic equation is

$$\lambda^2 + \frac{k}{m} = 0$$

whose solutions are

$$\lambda_1 = i\sqrt{k/m}, \quad \lambda_2 = -i\sqrt{k/m}$$

Solving Equation of Motion

$$\frac{d^2x}{dt^2} + \frac{k}{m}x = 0$$

$$\lambda_1 = i\sqrt{k/m}, \quad \lambda_2 = -i\sqrt{k/m}$$

Define $\omega = \sqrt{k/m}$; the general solution is

$$x(t) = c_1 \cos \omega t + c_2 \sin \omega t$$

Initial Conditions

$$x(t) = c_1 \cos \omega t + c_2 \sin \omega t$$

Assume that we know the conditions at $t = 0$:

$$x(0) = 0, \quad v(0) = \left. \frac{dx}{dt} \right|_{t=0} = v_0$$

From the first condition,

$$x(0) = c_1 \cos 0 + c_2 \sin 0 = c_1 = 0$$

$$x(t) = c_2 \sin \omega t$$

Initial Conditions

$$x(t) = c_2 \sin \omega t$$

$$v(0) = \left. \frac{dx}{dt} \right|_{t=0} = v_0$$

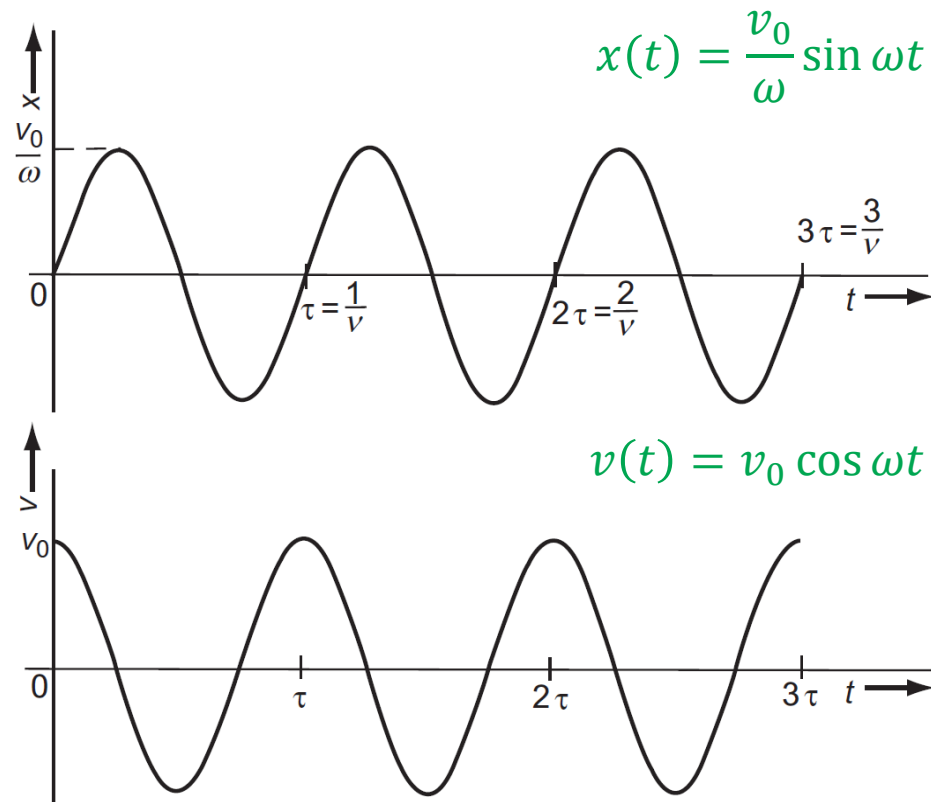
From the second condition,

$$v(0) = c_2 \omega \cos 0 = c_2 \omega = v_0$$

Hence, the solution is

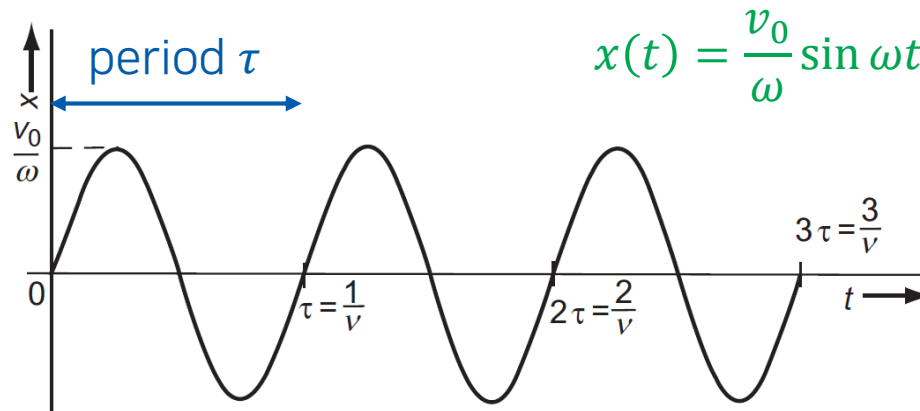
$$x(t) = \frac{v_0}{\omega} \sin \omega t$$

Uniform Harmonic Motion



(Image: Mortimer, *Mathematics for Physical Chemistry*, 4th ed., Figure 12.1)

Period and Frequency

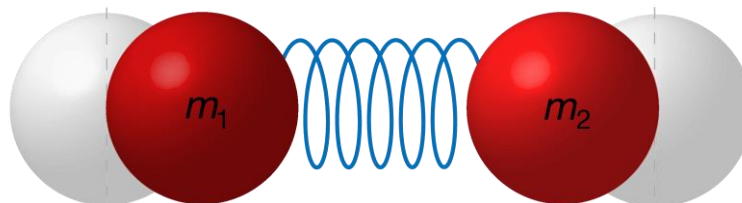


- Period (주기): $\omega\tau = 2\pi$

$$\tau = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{m}{k}}$$

- Frequency (진동수): $\nu = 1/\tau = \omega/2\pi$

Vibration of Diatomic Molecules



- Reduced mass

$$\mu = \frac{m_1 m_2}{m_1 + m_2}$$

- Vibrational frequency

$$\nu = \frac{\omega}{2\pi} = \frac{1}{2\pi} \sqrt{\frac{k}{\mu}}$$

(Image: <http://felix.chem.washington.edu/455/vibrations.html>)

Example 12.4

Calculate the frequency of vibration of a hydrogen fluoride molecule. The force constant k is equal to $966 \text{ N m}^{-1} = 966 \text{ J m}^{-2}$.

The reduced mass of the molecule is

$$\mu = \frac{m_1 m_2}{m_1 + m_2} = \frac{(1.008 \text{ g mol}^{-1})(18.998 \text{ g mol}^{-1})}{1.008 \text{ g mol}^{-1} + 18.998 \text{ g mol}^{-1}} = 0.9572 \text{ g mol}^{-1}$$

For one molecule,

$$\mu = 1.589 \times 10^{-27} \text{ kg}$$

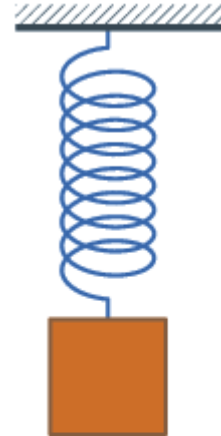
Hence, the vibrational frequency is

$$\nu = \frac{1}{2\pi} \sqrt{\frac{k}{\mu}} = \frac{1}{2\pi} \sqrt{\frac{966 \text{ kg s}^{-2}}{1.589 \times 10^{-27} \text{ kg}}} = 1.24 \times 10^{14} \text{ s}^{-1}$$

Damped Harmonic Oscillator

(감쇠 조화진동자)

- Damping (감쇠): reduction of oscillation
- Application of **friction force** (마찰력)
 - proportional to the velocity
 - in the opposite direction



$$\vec{F}_f = -\zeta \vec{v} = -\zeta \frac{d\vec{r}}{dt}$$

friction constant
(마찰 상수)

1-Dimensional Equation of Motion

$$F = m \frac{d^2 x}{dt^2} = -kx - \zeta \frac{dx}{dt}$$

- Standard form

$$\frac{d^2 x}{dt^2} + \frac{\zeta}{m} \frac{dx}{dt} + \frac{k}{m} x = 0$$

- Characteristic equation

$$\lambda^2 + \frac{\zeta}{m} \lambda + \frac{k}{m} = 0$$

Roots of Characteristic Equation

$$\lambda^2 + \frac{\zeta}{m}\lambda + \frac{k}{m} = 0$$

$$D = \left(\frac{\zeta}{m}\right)^2 - \frac{4k}{m}$$

1. Distinct real roots ($D > 0$)
2. Repeated real roots ($D = 0$)
3. Conjugate complex roots ($D < 0$)

(1) $D > 0$: Overdamping (과감쇠)

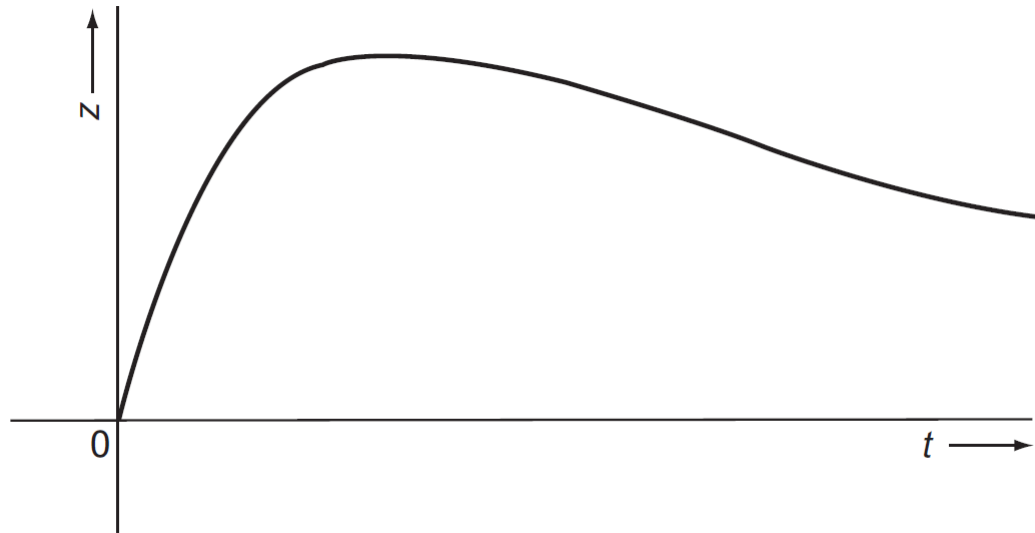
$$\lambda_1 = \frac{-\zeta + \sqrt{\zeta^2 - 4k}}{2m}, \quad \lambda_2 = \frac{-\zeta - \sqrt{\zeta^2 - 4k}}{2m}$$

$$x(t) = c_1 e^{\lambda_1 t} + c_2 e^{\lambda_2 t}$$

- Note that $\lambda_1, \lambda_2 < 0$ (Example 12.6) → decay!
- If the initial condition is $x(0) = 0$, $c_1 = -c_2$

(1) $D > 0$: Overdamping (과감쇠)

$$x(t) = c(e^{\lambda_1 t} - e^{\lambda_2 t})$$



(Image: Mortimer, *Mathematics for Physical Chemistry*, 4th ed., Figure 12.2)

(2) $D = 0$: Critical Damping (임계감쇠)

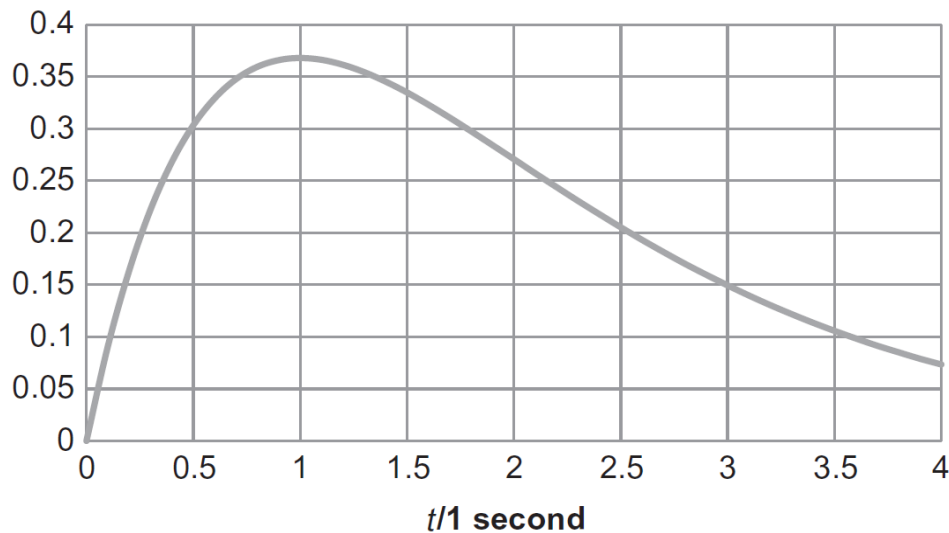
$$\lambda_1 = -\frac{\zeta}{2m}$$

$$x(t) = c_1 e^{\lambda_1 t} + c_2 t e^{\lambda_1 t}$$

- Note that $\lambda_1 < 0 \rightarrow$ decay!
- If the initial condition is $x(0) = 0$, $c_1 = 0$

(2) $D = 0$: Critical Damping (임계감쇠)

$$x(t) = c_2 t e^{\lambda_1 t}$$



(Image: Mortimer, *Mathematics for Physical Chemistry*, 4th ed., Figure 12.4)

Exercise 12.8

Substitute the trial solution $x(t) = c_1 e^{-\zeta t/2m} + c_2 t e^{-\zeta t/2m}$ into the equation of motion with the condition $\zeta^2 - 4mk = 0$, and show that the equation is satisfied.

The equation of motion is

$$\frac{d^2 x}{dt^2} + \frac{\zeta}{m} \frac{dx}{dt} + \frac{k}{m} x = 0$$

The derivatives of the trial solution are

$$\frac{dx}{dt} = c_1 \left(-\frac{\zeta}{2m} \right) e^{-\zeta t/2m} + c_2 e^{-\zeta t/2m} + c_2 \left(-\frac{\zeta}{2m} \right) t e^{-\zeta t/2m}$$

$$\frac{d^2 x}{dt^2} = c_1 \left(\frac{\zeta}{2m} \right)^2 e^{-\zeta t/2m} + 2c_2 \left(-\frac{\zeta}{2m} \right) e^{-\zeta t/2m} + c_2 \left(\frac{\zeta}{2m} \right)^2 t e^{-\zeta t/2m}$$

Exercise 12.8

Substitute the trial solution $x(t) = c_1 e^{-\zeta t/2m} + c_2 t e^{-\zeta t/2m}$ into the equation of motion with the condition $\zeta^2 - 4mk = 0$, and show that the equation is satisfied.

Substitution into the equation of motion leads to

$$c_1 \left(-\frac{\zeta^2}{4m^2} + \frac{4mk}{4m^2} \right) + c_2 t \left(-\frac{\zeta^2}{4m^2} + \frac{4mk}{4m^2} \right) = 0$$

The condition $\zeta^2 - 4mk = 0$ makes the left-handed side zero. Hence, the equation is satisfied.

(3) $D < 0$: Underdamping (미급감쇠)

$$\lambda_1 = \frac{-\zeta + \sqrt{\zeta^2 - 4k}}{2m}, \quad \lambda_2 = \frac{-\zeta - \sqrt{\zeta^2 - 4k}}{2m}$$

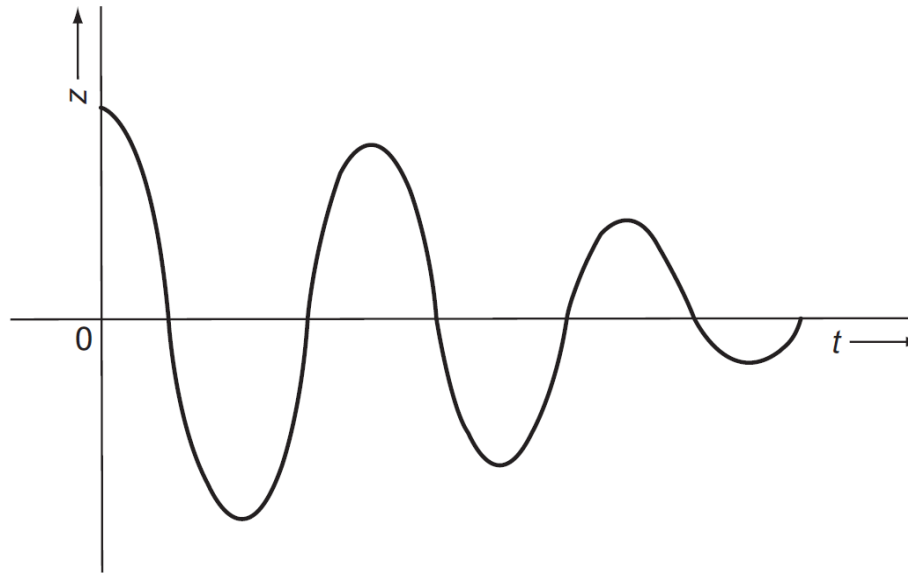
Let $\omega = \sqrt{(k/m) - (\zeta/2m)^2}$;

$$\lambda_1 = -\frac{\zeta}{2m} + i\omega, \quad \lambda_2 = -\frac{\zeta}{2m} - i\omega$$

$$x(t) = e^{-\zeta t/2m} (c_1 \cos \omega t + c_2 \sin \omega t)$$

(3) $D < 0$: Underdamping (미급감쇠)

$$x(t) = ce^{-\zeta t/2m} \cos \omega t$$



(Image: Mortimer, *Mathematics for Physical Chemistry*, 4th ed., Figure 12.3)

Inhomogeneous Linear ODE

$$a \frac{d^2 y}{dx^2} + b \frac{dy}{dx} + cy = \underline{g(x)}$$

inhomogeneous term
(비동차항)

- Complementary equation (보상방정식)

$$a \frac{d^2 y}{dx^2} + b \frac{dy}{dx} + cy = 0$$

Solving Inhomogeneous ODE

$$a \frac{d^2 y}{dx^2} + b \frac{dy}{dx} + cy = g(x)$$

1. Solve the complementary equation →
complementary function (보상함수) y_c
2. Find a particular solution y_p that satisfies the DE
3. The general solution is $y = y_c + y_p$

Example

Find the general solution to the differential equation

$$\frac{d^2y}{dx^2} - 5\frac{dy}{dx} + 4y = 16x.$$

A particular solution to this equation is $y = 4x + 5$.

The complementary equation is

$$\frac{d^2y}{dx^2} - 5\frac{dy}{dx} + 4y = 0$$

and the complementary function is

$$y_c(x) = c_1e^x + c_2e^{4x}$$

Hence, the general solution is

$$y = c_1e^x + c_2e^{4x} + 4x + 5$$

Method of Undetermined Coefficients

(미정계수법)

How can we obtain a particular solution y_p ?

Guess the form of y_p with undetermined coefficients!

Trial Solutions

Inhomogeneous term	Trial solution
1	A
t^n	$A_0 + A_1 t + A_2 t + \cdots + A_n t^n$
$e^{\alpha t}$	$A e^{\alpha t}$
$t^n e^{\alpha t}$	$e^{\alpha t} (A_0 + A_1 t + A_2 t + \cdots + A_n t^n)$
$e^{\alpha t} \sin(\beta t)$	$e^{\alpha t} [A \cos(\beta t) + B \sin(\beta t)]$
$e^{\alpha t} \cos(\beta t)$	$e^{\alpha t} [A \cos(\beta t) + B \sin(\beta t)]$

Example

Find the general solution to the differential equation

$$\frac{d^2y}{dx^2} + 4\frac{dy}{dx} + 4y = 2x^2 - 3x + 6.$$

The complementary equation is

$$\frac{d^2y}{dx^2} + 4\frac{dy}{dx} + 4y = 0$$

and the complementary function is

$$y_c(x) = c_1e^{-2x} + c_2xe^{-2x}$$

Example

Find the general solution to the differential equation

$$\frac{d^2y}{dx^2} + 4\frac{dy}{dx} + 4y = 2x^2 - 3x + 6.$$

$$y_c(x) = c_1e^{-2x} + c_2xe^{-2x}$$

Now, consider a trial solution $y_p = A_0 + A_1x + A_2x^2$, whose derivatives are

$$\frac{dy_p}{dx} = A_1 + 2A_2x, \quad \frac{d^2y_p}{dx^2} = 2A_2$$

Substitution leads to

$$4A_2x^2 + (8A_2 + 4A_1)x + (4A_0 + 4A_1 + 2A_2) = 2x^2 - 3x + 6$$

Comparison gives $A_2 = 1/2, A_1 = -7/4, A_0 = 3$.

Example

Find the general solution to the differential equation

$$\frac{d^2y}{dx^2} + 4\frac{dy}{dx} + 4y = 2x^2 - 3x + 6.$$

$$y_c(x) = c_1e^{-2x} + c_2xe^{-2x}$$

$$y_p(x) = \frac{1}{2}x^2 - \frac{7}{4}x + 3$$

The general solution is

$$y = c_1e^{2x} + c_2xe^{2x} + \frac{1}{2}x^2 - \frac{7}{4}x + 3$$

Glitch in the Method

A trial solution y_p can be
a solution of the complementary equation.

In this case, try $x^n y_p$ instead!

(n : smallest integer that resolves the problem)

Example

Find the general solution to the differential equation

$$\frac{d^2y}{dx^2} - 5\frac{dy}{dx} + 4y = 8e^x.$$

The complementary function is

$$y_c(x) = c_1e^x + c_2e^{4x}$$

According to Table 12.1, the trial solution for $8e^x$ is Ae^x .

However, substitution gives

$$Ae^x - 5Ae^x + 4Ae^x = 0 = 8e^x$$

The trial solution is a wrong guess.

Example

Find the general solution to the differential equation

$$\frac{d^2y}{dx^2} - 5\frac{dy}{dx} + 4y = 8e^x.$$

$$y_c(x) = c_1e^x + c_2e^{4x}$$

Instead of Ae^x , we can try $y_p = Axe^x$.

$$\frac{dy_p}{dx} = Ae^x + Axe^x, \quad \frac{d^2y_p}{dx^2} = 2Ae^x + Axe^x$$

Hence,

$$\frac{d^2y_p}{dx^2} - 5\frac{dy_p}{dx} + 4y_p = -3Ae^x$$

 and $A = -8/3$.

Example

Find the general solution to the differential equation

$$\frac{d^2y}{dx^2} - 5\frac{dy}{dx} + 4y = 8e^x.$$

$$y_c(x) = c_1e^x + c_2e^{4x}$$

$$y_p(x) = -\frac{8}{3}xe^x$$

Therefore, the general solution is

$$y(x) = c_1e^x + c_2e^{4x} - \frac{8}{3}xe^x$$